

Jose Barbeito
Swansea University MSc Personal Statement

I love understanding how things work. Whether it is a simple toy car, an intricate SLR camera, or something completely out of my wheelhouse like a SpaceX raptor engine, understanding the inner workings of a product or machine has always attracted me. As my engineering knowledge expanded, I began to appreciate the fundamental building blocks: electricity, thermodynamics, metal alloys, and semiconductors. I started to appreciate everyday objects not only for its components, but for the materials they were fabricated from. This new perspective opened 'a third eye' for me, and I was hooked.

At NC State, Materials Science showed me how fundamental choices at the atomic level shape entire industries. That realization is what drew me in—and why I want to continue pushing that knowledge forward through graduate study. My time at NC State has been defined by a constant desire to learn more about how materials shape our world. Whether it was defects in a crystal lattice or the complexities of nanomaterials, each course added another layer to the way I view everyday technologies.

Every material science class I have taken as an undergraduate has been fascinating. From Nanomaterials to Defects in Solids, every step I take into the world of Material Science deepens my interest in this field. In the future, I want to help push the boundary of human innovation in the material space, specifically in industry. I recently sat in one of my Material Science classes and heard a fellow student talk about how Material Science is the "jack of all trades, master of none." While often seen as a limitation, many forget the full passage: "Jack of all trades, master of none, is oftentimes better than master of one." Material Science gives a fundamental understanding of materials, detailing how they interact and influence the final product. This multidisciplinary nature is precisely why I believe a masters in Material Science is crucial for my career in industry and will allow me to push the boundaries of what is possible.

My personal work experience has been primarily in the semiconductor industry, working with silicon carbide as a part time employee at Wolfspeed. Being able to understand the crystal structure of the boule, how the lattice affects the method of cutting and separation of the wafer, and the physical properties of chemical vapor deposition in the epitaxial growth process has allowed me to quickly integrate into the team and become a valuable asset. I hope that my time in graduate school will allow me to expand on that knowledge, allowing me to understand these processes better and to develop new processes to increase efficiency and yield, while lowering cost. Eventually I want to transition into the crystal growth department, in which understanding the material properties will be crucial.

What excites me about Swansea University's Materials Engineering with Industry MSc is the direct collaboration with industry partners and the strong research backbone. The opportunity to apply my knowledge in real-world contexts while continuing to learn aligns perfectly with my goals. I'm especially drawn to the program's emphasis on advanced materials and their applications across high-performance industries.

By continuing my education at Swansea University, I hope to become part of the select few who push the boundaries in my chosen field. I hope that the knowledge gained from this masters program will help me accomplish that.